

1. USV and controller parameters

First, given the matrices A_η , A_v and B_{uw} as:

$$\begin{aligned} A_\eta &= -M^{-1}G = \begin{bmatrix} -0.0358 & 0 & 0 \\ 0 & -0.0208 & 0 \\ 0 & -0.0394 & 0 \end{bmatrix}, \\ A_v &= -M^{-1}N = \begin{bmatrix} -0.0797 & 0 & 0 \\ 0 & -0.0818 & -0.0577 \\ 0 & -0.2254 & -0.2535 \end{bmatrix}, \\ B_{uw} &= M^{-1} = \begin{bmatrix} 0.9215 & 0 & 0 \\ 0 & 0.7802 & 1.4811 \\ 0 & 1.4811 & 7.4562 \end{bmatrix}. \end{aligned}$$

Accordingly, for $\theta(t) \in [-\frac{\pi}{6}, \frac{\pi}{6}]$, the nonlinear USV can be modeled by the subsequent local T-S fuzzy systems:

$$\begin{aligned} A_i &= \begin{bmatrix} 0 & J_i \\ A_\eta & A_v \end{bmatrix}, \quad B_i = \begin{bmatrix} 0 \\ B_{uw} \end{bmatrix}, \\ C_{2i} &= [0 \quad 0 \quad -0.002 \quad 0 \quad 0 \quad 0.01], i \in \{1, 2, 3, 4\} \end{aligned}$$

where the corresponding rotational matrices are defined as:

$$\begin{aligned} J_1 &= \begin{bmatrix} 1 & -1/2 & 0 \\ 1/2 & -1/2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, J_2 = \begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 \\ 1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ J_3 &= \begin{bmatrix} 1 & 1/2 & 0 \\ -1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, J_4 = \begin{bmatrix} \sqrt{3}/2 & 1/2 & 0 \\ -1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

Furthermore, the FMF are explicitly constructed as:

$$\begin{aligned} g_1(\theta(t)) &= \phi_1(x_3(t))(-3 - 2\sqrt{3} + (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_2(\theta(t)) &= \phi_1(x_3(t))(4 + 2\sqrt{3} - (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_3(\theta(t)) &= \phi_2(x_3(t))(-3 - 2\sqrt{3} + (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_4(\theta(t)) &= \phi_2(x_3(t))(4 + 2\sqrt{3} - (4 + 2\sqrt{3})\cos(x_3(t))), \end{aligned}$$

in which the intermediate variables ϕ_1 and ϕ_2 are given by:

$$\phi_1(x_3(t)) = \frac{1}{2} + \sin(x_3(t)), \quad \phi_2(x_3(t)) = \frac{1}{2} - \sin(x_3(t)).$$

Meanwhile, the time-varying environmental disturbances $w(t)$ are described as:

$$\begin{cases} w_1(t) = -0.1e^{-t} \\ w_2(t) = -0.4\cos(1.58t)e^{-0.235t} \\ w_3(t) = -0.2\sin(2t)e^{-2t} \end{cases} \quad (1)$$

The feedback control gains alongside the triggering weight matrices are computed as follows:

$$\begin{aligned} K_{11} &= \begin{bmatrix} -0.2878 & -0.0043 & 0.0000 & -0.8694 & 0.0000 & 0.0003 \\ 0.0079 & -0.5411 & 0.1138 & -0.0005 & -1.6602 & 0.3208 \\ -0.0012 & 0.1024 & -0.0748 & 0.0005 & 0.3329 & -0.1614 \end{bmatrix}, \\ K_{12} &= \begin{bmatrix} -0.1691 & -0.0024 & 0.0000 & -0.5111 & 0.0002 & 0.0002 \\ 0.0045 & -0.3171 & 0.0708 & -0.0006 & -0.9748 & 0.1965 \\ -0.0007 & 0.0597 & -0.0478 & 0.0003 & 0.1953 & -0.1027 \end{bmatrix}, \end{aligned}$$

$$K_{13} = \begin{bmatrix} -0.0877 & -0.0012 & 0.0000 & -0.2654 & 0.0002 & 0.0001 \\ 0.0022 & -0.1643 & 0.0384 & -0.0005 & -0.5058 & 0.1054 \\ -0.0004 & 0.0308 & -0.0265 & 0.0002 & 0.1012 & -0.0567 \end{bmatrix},$$

$$K_{21} = \begin{bmatrix} -0.2929 & -0.0057 & -0.0000 & -0.8840 & 0.0007 & 0.0002 \\ 0.0099 & -0.5496 & 0.1133 & -0.0019 & -1.6831 & 0.3211 \\ -0.0015 & 0.1041 & -0.0747 & 0.0007 & 0.3366 & -0.1614 \end{bmatrix},$$

$$K_{22} = \begin{bmatrix} -0.1721 & -0.0031 & 0.0000 & -0.5196 & 0.0007 & 0.0002 \\ 0.0055 & -0.3221 & 0.0707 & -0.0016 & -0.9881 & 0.1972 \\ -0.0009 & 0.0606 & -0.0478 & 0.0005 & 0.1975 & -0.1028 \end{bmatrix},$$

$$K_{23} = \begin{bmatrix} -0.0892 & -0.0015 & 0.0000 & -0.2695 & 0.0007 & 0.0001 \\ 0.0026 & -0.1666 & 0.0385 & -0.0013 & -0.5118 & 0.1059 \\ -0.0004 & 0.0312 & -0.0265 & 0.0003 & 0.1022 & -0.0568 \end{bmatrix},$$

$$K_{31} = \begin{bmatrix} -0.2878 & 0.0043 & -0.0000 & -0.8694 & -0.0000 & -0.0003 \\ -0.0079 & -0.5411 & 0.1138 & 0.0005 & -1.6602 & 0.3208 \\ 0.0012 & 0.1024 & -0.0748 & -0.0005 & 0.3329 & -0.1614 \end{bmatrix},$$

$$K_{32} = \begin{bmatrix} -0.1691 & 0.0024 & -0.0000 & -0.5111 & -0.0002 & -0.0002 \\ -0.0045 & -0.3171 & 0.0708 & 0.0006 & -0.9748 & 0.1965 \\ 0.0007 & 0.0597 & -0.0478 & -0.0003 & 0.1953 & -0.1027 \end{bmatrix},$$

$$K_{33} = \begin{bmatrix} -0.0877 & 0.0012 & -0.0000 & -0.2654 & -0.0002 & -0.0001 \\ -0.0022 & -0.1643 & 0.0384 & 0.0005 & -0.5058 & 0.1054 \\ 0.0004 & 0.0308 & -0.0265 & -0.0002 & 0.1012 & -0.0567 \end{bmatrix},$$

$$K_{41} = \begin{bmatrix} -0.2929 & 0.0057 & 0.0000 & -0.8840 & -0.0007 & -0.0002 \\ -0.0099 & -0.5496 & 0.1133 & 0.0019 & -1.6831 & 0.3211 \\ 0.0015 & 0.1041 & -0.0747 & -0.0007 & 0.3366 & -0.1614 \end{bmatrix},$$

$$K_{42} = \begin{bmatrix} -0.1721 & 0.0031 & 0.0000 & -0.5196 & -0.0007 & -0.0002 \\ -0.0055 & -0.3221 & 0.0707 & 0.0016 & -0.9881 & 0.1972 \\ 0.0009 & 0.0606 & -0.0478 & -0.0005 & 0.1975 & -0.1028 \end{bmatrix},$$

$$K_{43} = \begin{bmatrix} -0.0892 & 0.0015 & 0.0000 & -0.2695 & -0.0007 & -0.0001 \\ -0.0026 & -0.1666 & 0.0385 & 0.0013 & -0.5118 & 0.1059 \\ 0.0004 & 0.0312 & -0.0265 & -0.0003 & 0.1022 & -0.0568 \end{bmatrix}.$$

$$\Psi_1 = \begin{bmatrix} 0.3464 & -0.0004 & -0.0006 & 1.0078 & 0.0088 & -0.0003 \\ -0.0004 & 0.3657 & -0.0089 & -0.0083 & 1.0632 & -0.0854 \\ -0.0006 & -0.0089 & 0.1347 & -0.0004 & -0.0710 & 0.2503 \\ 1.0078 & -0.0083 & -0.0004 & 3.0651 & 0.0003 & -0.0004 \\ 0.0088 & 1.0632 & -0.0710 & 0.0003 & 3.2362 & -0.3371 \\ -0.0003 & -0.0854 & 0.2503 & -0.0004 & -0.3371 & 0.5005 \end{bmatrix},$$

$$\Psi_2 = \begin{bmatrix} 0.3499 & -0.0003 & -0.0006 & 1.0356 & 0.0093 & -0.0002 \\ -0.0003 & 0.3686 & -0.0088 & -0.0089 & 1.0890 & -0.0864 \\ -0.0006 & -0.0088 & 0.1347 & -0.0004 & -0.0703 & 0.2503 \\ 1.0356 & -0.0089 & -0.0004 & 3.1345 & 0.0008 & -0.0003 \\ 0.0093 & 1.0890 & -0.0703 & 0.0008 & 3.2977 & -0.3401 \\ -0.0002 & -0.0864 & 0.2503 & -0.0003 & -0.3401 & 0.5007 \end{bmatrix},$$

$$\Psi_3 = \begin{bmatrix} 0.3464 & 0.0004 & 0.0006 & 1.0078 & -0.0088 & 0.0003 \\ 0.0004 & 0.3657 & -0.0089 & 0.0083 & 1.0632 & -0.0854 \\ 0.0006 & -0.0089 & 0.1347 & 0.0004 & -0.0710 & 0.2503 \\ 1.0078 & 0.0083 & 0.0004 & 3.0651 & -0.0003 & 0.0004 \\ -0.0088 & 1.0632 & -0.0710 & -0.0003 & 3.2362 & -0.3371 \\ 0.0003 & -0.0854 & 0.2503 & 0.0004 & -0.3371 & 0.5005 \end{bmatrix},$$

$$\Psi_4 = \begin{bmatrix} 0.3499 & 0.0003 & 0.0006 & 1.0356 & -0.0093 & 0.0002 \\ 0.0003 & 0.3686 & -0.0088 & 0.0089 & 1.0890 & -0.0864 \\ 0.0006 & -0.0088 & 0.1347 & 0.0004 & -0.0703 & 0.2503 \\ 1.0356 & 0.0089 & 0.0004 & 3.1345 & -0.0008 & 0.0003 \\ -0.0093 & 1.0890 & -0.0703 & -0.0008 & 3.2977 & -0.3401 \\ 0.0002 & -0.0864 & 0.2503 & 0.0003 & -0.3401 & 0.5007 \end{bmatrix}.$$